

Regulation Management: Pain Points & AI Impact

Where effort goes today, what AI-assisted automation removes, and what should stay human

Quantified in days & person-effort, not dollars. Money follows from time; time is a claim we can defend and a judge can sanity-check.

Automation is a boundary, not a blanket. Some steps should stay human — that boundary is a feature of the design, not a gap in it.

Maps to the Human + AI Work, Enterprise Future, and Responsible AI pillars.

Pain Points → AI-Assisted Effort Reduction (Stages 1–4)

Stage	Pain Point Today	Current Effort	With AI-Assisted Platform	Stays Human — Why
1. Intake	Signals scattered across email, portals, bulletins; easy to miss or catch late	2–4 hrs/week per function, reactive	Continuous structured feed, near-zero monitoring effort	—
2. Decomposition	Long legal text; broken into actionable clauses ad hoc, once, by one SME	1–2 days per regulation, not reusable	Minutes; machine-readable and reusable downstream	SME validates ambiguous clauses
3. Applicability	SME-by-SME consultation; judgment never recorded reusably	3–5 days elapsed (mostly waiting), 4–8 hrs actual SME time	AI proposal in minutes; SME reviews same day	Always — core judgment call, never silent-automated
4. Impact Fan-Out	Manual BOM/system traversal; each function re-learns the regulation cold	5–10 days elapsed, 15–20 person-hrs combined	AI traversal in hours; meetings start from a pre-scoped brief	Cross-functional trade-off negotiation stays a meeting

Effort figures are SME-reasoned estimates for a synthetic scenario, not measured Cummins data — stated as directional, not precise.

Pain Points → AI-Assisted Effort Reduction (Stages 5–8 + Amendment Path)

Stage	Pain Point Today	Current Effort	With AI-Assisted Platform	Stays Human — Why
5. Exposure Quantification	Estimated informally, inconsistently, often skipped under pressure	2–3 days, inconsistent method	Minutes; consistent, precedent-referenced	Final materiality judgment stays human
6. Action Plan	Built per-function in spreadsheets, rarely linked back to source clause	2–4 days	Minutes; structured, one action per obligation	Resourcing trade-offs vs. other priorities
7. Governance Decision	Scattered across meetings, emails, decks; no single retrievable record	1–2 weeks calendar time, mostly scheduling delay	Async structured review, days not weeks	Always — approval stays a named human, permanently
8. Handoff & Evidence	Evidence lives in inboxes/drives; audit readiness = manual reconstruction	Days of reconstruction, per audit request	One click; always current	Sufficiency judgment on borderline evidence
Amendment Path	Amendments missed, or full process restarted from scratch	Full re-cycle, ~2–3 weeks, or missed entirely	Auto-detected; blast radius in minutes; only changed items reach a human	Always — closing/re-opening work is never auto-executed

Effort figures are SME-reasoned estimates for a synthetic scenario, not measured Cummins data — stated as directional, not precise.

What Stays Human — And Why

We didn't automate judgment. We automated the waiting, the searching, and the reconstructing around it.

H

Applicability Rulings

AI proposes with confidence + citation. SME always decides — never silently automated.

H

Governance Approval

Every gate is a named human, permanently. This is the audit trail's anchor.

H

Amendment Disposition

Detection and blast-radius are automatic. Closing or re-opening a task is never auto-executed.

H

Cross-Functional Negotiation

The meeting stays. Only the cold-start research before it goes away.

H

Evidence Sufficiency (borderline cases)

AI flags gaps. A human rules on adequacy when it's ambiguous.

The Compliance Analyst's Role Shifts — It Doesn't Disappear

The platform removes the parts of the job that were never really about compliance expertise — **the searching, the waiting, the reconstructing** — and concentrates their time on the parts that were: applicability judgment, risk materiality, cross-functional negotiation, and evidence sufficiency.

How This Maps to the Judging Pillars

Human + AI Work

The stage-by-stage boundary above IS the roles/reskilling model — shown with reasoning, not asserted in the abstract.

Enterprise Future

The effort-reduction deltas per stage feed directly into the value-realization story.

Responsible AI

The “stays human” list is the same control boundary, restated in workforce terms — one design, two pillars.

Obligation Deduplication & Overlap Detection

Sits at the seam between Stage 2 (Decomposition) and Stage 3 (Applicability) — reused by the Amendment diff step too

TIER A — Deduplication

Mechanical match against the existing obligation library — same requirement re-issued or restated.

Confidence: High

Action: Auto-flag “Already Tracked,” link to existing record, skip redundant applicability work.

TIER B — Overlap Detection

Semantic overlap across different regulators/jurisdictions — same underlying requirement, different threshold or test method.

Confidence: Lower — always human-gated

Action: Route to SME with the related obligation + evidence pre-attached as an equivalence question, never auto-resolved.

What we do NOT claim: “Existing regulation covers the new regulation.” That's a legal-scope claim we never make.

What we DO claim: A technical parameter in your certified product data may already satisfy a new obligation's threshold — verify equivalence, then reuse evidence instead of re-engineering.

WOW Factor Priority Map

Cross-cutting capabilities layered onto the existing two flows — not new paths

Feature	Build Effort	Where It Hooks In	Priority
Obligation Dedup & Overlap Detection	Low — extends Stage 2 output	Stage 2 → 3 seam, both flows	BUILD DEEP — Hero #1
Conversational Query Layer (NL over the graph)	Low — reads existing data, no new logic	Overlay across all screens	BUILD IF TIME — Hero #2
Regulatory Horizon / Anticipatory View	Very low — uses lifecycle field already planned	Dashboard overlay	One extra screen
AI Divergence / Self-Audit Dashboard	Very low — reads existing Decision log	Reads Decision records	One screen / slide mention

Design rule: no third pipeline path. Every WOW factor above is a layer over the existing 8-stage and 5-stage flows — same seed data, same state machine, no added surface to break live.

Same Engine, Swapped Config — Portability Beat

A 90-second live demo beat, run right after the manufacturing walkthrough



Fixed — identical for every industry

Config Element	Manufacturing	Banking
Scope dimensions	Product line × plant × market	Legal entity × business line × jurisdiction
Impact taxonomy source	BOM / engineering spec traversal	Product / process / core-system traversal
Regulator / evidence source	EPA, ECHA, ISO	OCC, FinCEN, FINRA
Sample obligation	MY2027 NOx threshold, 0.035 g/hp-hr	Updated KYC beneficial-ownership threshold

Live beat script: “Everything you've seen ran on this same eight-stage engine. Toggle to Banking — same intake, same decomposition, same governance gate. What changed: scope, impact traversal, and regulator source. Same engine. Swapped config.”

AI Divergence Log — Live Accountability

Not a background table — shown live on screen, updating as recommendations are accepted or overridden

Total recommendations	Human overrides	Agreement rate
3	1	67%
MY2030 useful-life extension, engine family X		Match
AI: Applicable (91% confidence) Human: Agrees		
PFAS sealing-application derogation		Overridden
AI: Not applicable (58% confidence) Human: Overrides → requires investigation		
Overlap: EU vs export NOx threshold		Match
AI: Overlap — verify equivalence (74% confidence) Human: Agrees, after review		